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Abbreviations

1. NLP - Natural Language processing
2. ANN – Artificial Neural Networks
3. Xg boost - eXtreme Gradient Boost
4. DBN - Deep Belief Neural
5. CNN - Convolutional Neural Network
6. RNN - Recurrent Neural Network
7. BOW – Bag of words.
8. TD-IDF - Term Frequency - Inverse Document Frequency
9. ML – Machine Learning

Abstract

In the world of digital transformation, online reviews play a major influence over consumer decision-making processes, particularly within the hospitality industry. It is imperative for hotels to comprehend the sentiments conveyed in these reviews to refine guest experiences and maintain a competitive edge. This project aims to conduct sentiment analysis on hotel reviews sourced from Tripadvisor, a prominent online platform renowned for travel reviews. Leveraging natural language processing (NLP) techniques and machine learning algorithms, we extract meaningful insights from textual data to categorize reviews as either positive or negative, thereby offering valuable feedback to hotel management. The study encompasses various stages, including data collection, preprocessing, and exploratory data analysis (EDA) to gain a comprehensive understanding of review characteristics. Subsequently, sentiment analysis models such as machine learning and neural networks are employed for review classification. The project culminates in the provision of actionable insights to hotels, enabling them to enhance service quality, address customer concerns, and elevate guest satisfaction levels.

Chapter 1

INTRODUCTION

1.1 Background

TripAdvisor stands out as a prominent platform for global travellers, providing essential guidance and evaluations on lodging, dining, and attractions. Its extensive user base contributes diverse experiences and opinions, establishing it as a dependable resource for travel planning. Hotel reviews, a central feature of the platform, offer insights, feedback, and suggestions from guests, aiding others in their travel arrangements. With a wealth of user-generated content, TripAdvisor has transformed the way individuals research and reserve accommodations, becoming an indispensable tool in modern travel. As one of the leading online travel platforms, TripAdvisor boasts over hundreds of millions of reviews and opinions spanning accommodations, restaurants, attractions, and more.

Founded in 2000, the platform has evolved into a trusted community-driven hub where travellers share their firsthand experiences to help others make informed decisions.

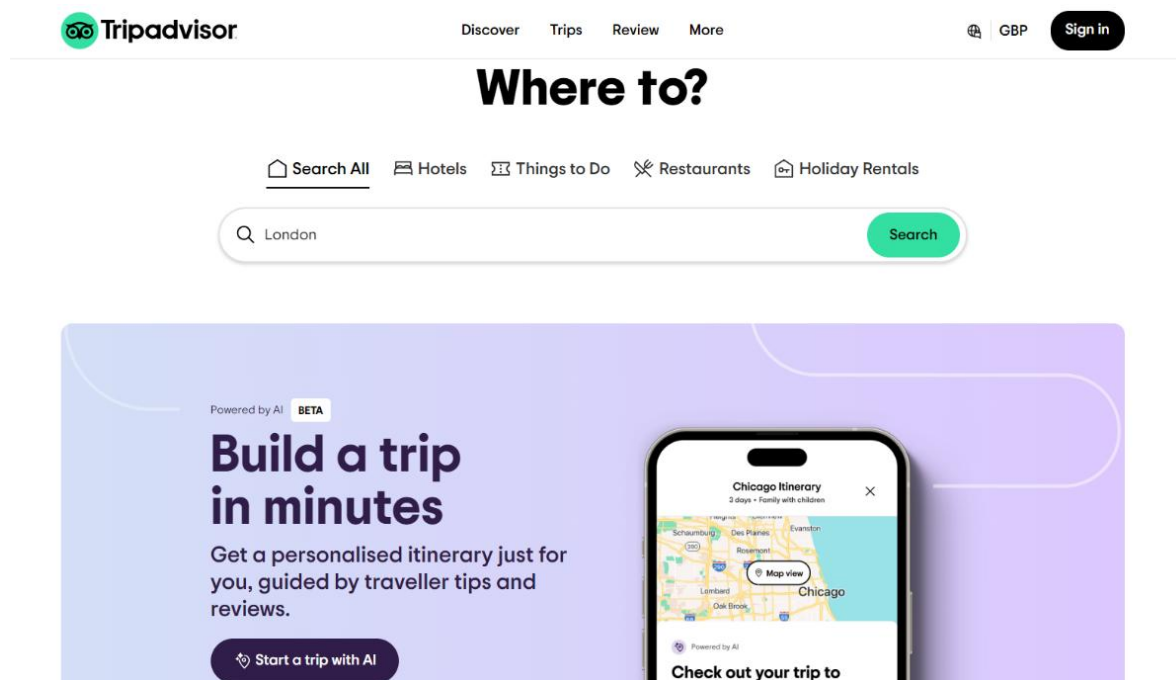


Figure 1: Homepage image (Trip Advisor, n.d.).

Since its establishment in 2000, TripAdvisor has evolved into a community-driven hub where travellers share firsthand experiences to aid others in decision-making. Hotel reviews cover a broad spectrum of properties, catering to various budgets and destinations worldwide. They provide prospective travellers with a glimpse into the guest experience, aiding in informed decision-making when selecting accommodations.

On most hotel review websites, they ask travellers to rate hotels in two ways: by giving them a star rating from one to five, and by writing a description of their experience. Traveller's ratings can differ depending on whether the hotel is independently owned or part of a chain. Independent hotels are properties that are privately owned and not associated with any specific brand. On the other hand, chain hotels are part of a larger hotel group or consortium. When it comes to rating patterns, it is important to note that independent and chain hotels may not always be directly comparable due to their differences.

Owing to their differences rating trends drawn by chain and independent hotels might not always be equivalent (Snehasish Banerjee, 2016). Currently, more than 200 million reviews appear across numerous travel-related review sites (Boo, 2018).

TripAdvisor's offerings are hotel reviews, a treasure trove of firsthand accounts, ratings, and recommendations from guests who have experienced accommodations firsthand. These reviews serve as a virtual window into the guest experience, offering invaluable insights into the quality of service, amenities, cleanliness, and overall ambiance of a hotel.

TripAdvisor's hotel reviews serve as a virtual window into the quality of service, amenities, cleanliness, and overall atmosphere of a hotel. They not only offer convenience but also influence traveller's perceptions and booking choices.

Positive reviews serve as endorsements, fostering confidence in potential guests, while negative reviews can deter them, signalling areas for improvement.

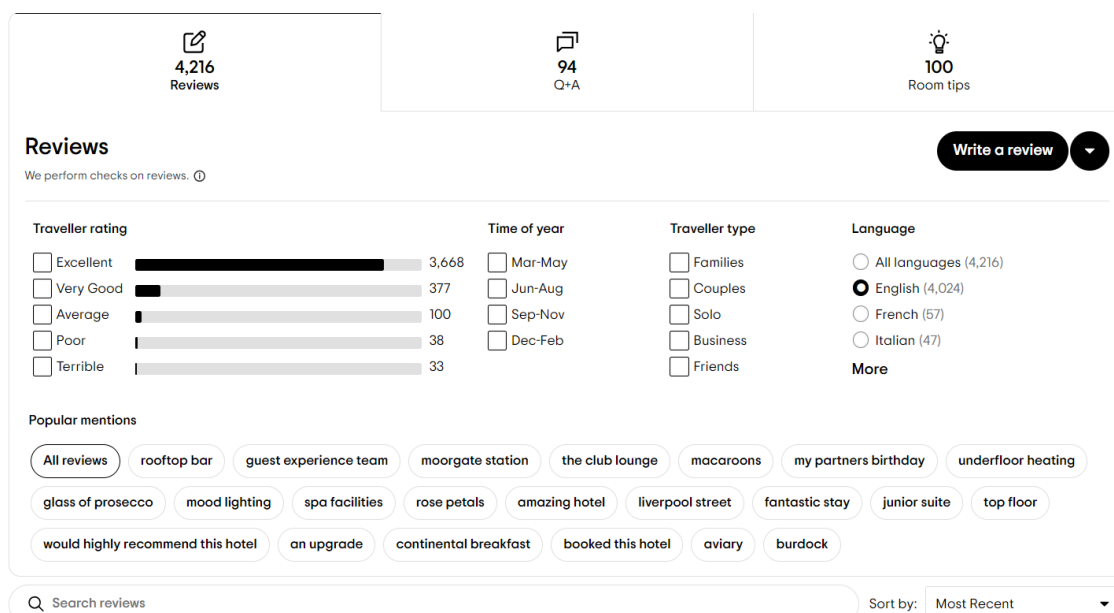


Figure 2: Review page image (Trip Advisor, n.d.).

For hoteliers, TripAdvisor reviews present both opportunities and challenges. Positive reviews can boost a property's reputation, attract new guests, and foster loyalty. Conversely, negative reviews can harm credibility and revenue, highlighting areas needing attention.

The appeal of TripAdvisor reviews lies in their authenticity and impartiality. They are authored by real travellers with no vested interests, offering unbiased assessments that others can rely on or lodging decisions.

The screenshot shows a TripAdvisor review interface. At the top, a user profile for Philip C is displayed, indicating he wrote a review in February 2024 and has 2 contributions with 3 helpful votes. The review itself is a 5-star rating, marked as a 'Hotel's Favourite'. The title is 'Top Team At Reception'. The text of the review praises the reception team and the hotel's location near Liverpool St and the Elizabeth line. Below the review, the date of stay is listed as January 2024, and the trip type is 'Travelled on business'. There are also star ratings for Value, Service, and Location. A disclaimer at the bottom states that the review is a subjective opinion. Below the review, there is a response from Deepak Panicker, the Hotel Manager, dated 3 February 2024, thanking Philip for his feedback.

Philip C wrote a review Feb 2024
2 contributions • 3 helpful votes

★★★★★ Hotel's Favourite

Top Team At Reception

The Reception team were particularly helpful and although I did not have the opportunity to use all the facilities I was happy with my stay. the location of the hotel is really great and very close to Liverpool St and the Elizabeth line. the investment and development in the area, with a mass variety of eateries etc makes this a great location to enjoy London.

Read less ▲

Date of stay: January 2024

Trip type: Travelled on business

★★★★★ Value
★★★★★ Service
★★★★★ Location

This review is the subjective opinion of a Tripadvisor member and not of Tripadvisor LLC. Tripadvisor performs checks on reviews as part of our industry-leading trust & safety standards. Read our [transparency report](#) to learn more.

👍 0

Response from Deepak Panicker- Hotel Manager, General Manager at The Montcalm Royal London House
Responded 3 Feb 2024

Dear Philip.

Thank you so much for your marvellous feedback.

We truly appreciate your feedback and value your business in every possible manner.

Figure 3: Review page (Trip Advisor, n.d.).

But what prompts travellers to take the time to share their experiences on TripAdvisor? And how do these reviews influence the decisions of fellow travellers? To answer these questions, we dive into the psychology behind reviewing behaviour, examining the motivations, emotions, and cognitive processes that drive individuals to contribute to the platform.

Understanding the psychology behind reviewing behaviour is crucial in comprehending why travellers share their experiences on TripAdvisor and how these reviews influence others. Through an exploration of motivations, emotions, and cognitive processes, this report aims to uncover insights into the significance of TripAdvisor hotel reviews, their impact on decision-making, and implications for the hospitality industry. By analysing trends and patterns,

stakeholders can gain valuable insights to enhance guest satisfaction, manage online reputation, and stay competitive in the evolving hospitality landscape.

1.2 Problem Statement

In the hospitality sector, grasping customer sentiment holds significant importance for both hotel managers and travellers. With the rapid expansion of online review platforms such as TripAdvisor, evaluating sentiment in hotel reviews has become vital for maintaining and enhancing service standards, as well as aiding travellers in making well-informed choices (Mukesh A. Zaveri, August 2011). However, the vast quantity of reviews presents a considerable challenge for manual analysis. Our focus is on addressing two fundamental issues:

Sentiment Analysis: We aim to create an efficient sentiment analysis model tailored specifically to TripAdvisor hotel reviews. This model should accurately categorize reviews as positive, negative, or neutral based on the sentiments expressed by reviewers (Pooja Mehta, p. February 2020). By automating this process, hotel managers can gain valuable insights into guest satisfaction levels, pinpoint areas for enhancement, and promptly address negative feedback to mitigate potential damage to their reputation.

Utilizing Sentiment Analysis for Hotel Improvement: The second aspect of our problem involves leveraging insights obtained from sentiment analysis to drive improvements in hotel services and guest experiences. Hotel managers require actionable feedback from reviews to identify areas for enhancement, address guest concerns, and enhance the quality of their establishments. However, the sheer volume of reviews on platforms like TripAdvisor poses a scalability challenge in efficiently extracting and prioritizing actionable insights.

Our research aims to develop innovative solutions to these challenges by leveraging techniques such as natural language processing (NLP), machine learning algorithms, and data mining methodologies. By offering accurate sentiment analysis and enhancing the assessment of review quality, our objective is to empower both travellers and hotel managers with the necessary tools to effectively navigate the intricate realm of online hotel reviews.

1.3 Justification of the study

The dataset utilized in this research comprises over 50,000 customer reviews and ratings for 142 luxury hotels across Europe. Europe, being a well-established tourism market with its natural landscapes and historic attractions, annually draws millions of tourists. Given the advancements in technology and evolving market dynamics, it is imperative for managers within the European luxury hotel industry to stay abreast of these developments. An

exploratory examination of this sector in Europe can provide valuable insights to aid hotel managers in making informed marketing decisions.

The primary justifications are below:

Enhanced Decision Making for Travellers: TripAdvisor hotel reviews serve as pivotal resources for travellers in selecting accommodations. By accurately assessing the sentiments conveyed in these reviews, travellers can make more informed decisions that align with their preferences and expectations. A robust sentiment analysis model can assist in navigating the extensive volume of reviews, thus saving travellers time and effort in their decision-making process.

Improvement of Service Quality: For hotel managers, comprehending the sentiments expressed in guest reviews is crucial for identifying areas of strength and areas requiring improvement. By analysing sentiments across various aspects of the guest experience, such as cleanliness, staff interactions, and amenities, hotel managers can pinpoint specific areas necessitating enhancements to elevate guest satisfaction and loyalty.

Competitive Advantage: Hotels consistently receiving positive sentiments in their reviews may gain a competitive edge over counterparts. Favourable reviews can bolster a hotel's reputation, attract more guests, and lead to increased revenue. Conversely, negative sentiments indicate areas for improvement, presenting an opportunity for hotels to proactively address issues and mitigate potential harm to their reputation.

Insights into Consumer Preferences: Sentiment analysis of TripAdvisor hotel reviews offers valuable insights into consumer preferences and trends within the hospitality industry. By analysing sentiments over time and across different demographic segments, researchers and industry analysts can identify emerging patterns, preferences, and areas for innovation and development within the hotel sector.

Data-Driven Decision Making: In an era prioritizing data-driven decision-making, sentiment analysis of hotel reviews provides a quantitative approach to understanding guest feedback. By quantifying sentiments and aggregating them at scale, hotel managers can prioritize actions and allocate resources effectively to address concerns and leverage strengths.

The dataset was sourced from Kaggle.com, being a prominent platform for hospitality information, possesses a robust review and rating system that efficiently collects customer feedback. This system enables the rapid collection of a large volume of customer reviews from hotels worldwide in a streamlined manner, benefiting customers, hotel managers, and academics alike.

Each observation in the dataset represents a single customer review for a particular hotel. These reviews consist of text detailing the customer's experience at the hotel, along with the corresponding rating provided by the reviewer and the nationality of the reviewer.

This study of sentiment analysis in TripAdvisor hotel reviews contributes to a deeper comprehension of guest perceptions, preferences, and behaviours, thereby facilitating improvements in service quality, customer satisfaction, and overall competitiveness within

the hospitality industry. Achieve classification, an exhaustive review of existing machine learning models in the literature will be conducted. The model demonstrating the highest performance will then be adopted and refined, or a hybrid machine learning approach will be employed.

1.4 Aim and Objective

In this project, we aim to investigate Trip Advisor hotel reviews and their associated ratings, reflecting customer experiences. Our approach involves leveraging Natural Language Processing (NLP) for extracting features and preprocessing data to prepare it for machine learning models. The primary objective is to develop sentiment analysis capabilities and predict ratings for Trip Advisor reviews. Through the application of NLP techniques and machine learning algorithms such as Naive Bayes, Logistic Regression, XG Boost classifier, and neural network models, our system seeks to categorize reviews into positive, neutral, or negative sentiments. On the other hand, sentiment score relies on one of two models. Those models are Bag-of-words model (BOW) and TF-IDF model (Hussein, JUNE 2016). The overarching aim is to offer valuable insights to both travellers and businesses by automating the analysis of vast quantities of Trip Advisor reviews.

Chapter 2

Literature Review

2.1 Sentiment Analysis

Sentiment analysis, which is also referred to as opinion mining, is a technique in natural language processing (NLP) that is used to identify the sentiment conveyed in a text. The goal of the sentiment analysis is to determine the emotion behind the text. When we analyse and categorise the information in the text, we determine them based on the polarity (positive-negative) classification to 3-class (positive -negative-neutral) classification, by taking the neutral feelings into consideration (Patil, December 2015). The main objective of sentiment analysis is to comprehend the general sentiment or viewpoint conveyed by the author regarding a specific subject, product, service, or topic.

Recently, sentiment analysis has become widely accepted, not only by researchers but also by businesses, governments, and organisations. The Internet has become incredibly popular and is now considered the main source of information worldwide (Wankhade, 2022). Most of the people rely on different online platforms to share their thoughts and opinions. Keep a close eye on public opinion and help with decision-making, we need to use data generated by users and analyse it automatically. Recently, sentiment analysis has become more popular among research communities (Sánchez-Rada, 2019).

2.2 Impact of sentiment analysis in hospitality

In the hospitality industry, sentiment analysis has become important. It helps businesses understand how customers feel and what they think, which is crucial for success. Hospitality businesses like hotels, restaurants, and travel agencies get a lot of feedback from their guests and customers. User-generated contents are considered today as a major source of information, and this is more helpful to industries like tourism where consumers need information that is unbiased and unregulated. These online text reviews are one such source of information that tourists find reliable and trustworthy (Sudhakar, 2019). This feedback comes from various sources like online reviews, surveys, and social media.

Businesses can improve the overall guest experience by analysing customer feedback and using it to identify areas of strength and weakness in their services. This allows them to make data-driven decisions to enhance their services. When it comes to making booking decisions, online review platforms such as TripAdvisor, Yelp, and Google Reviews have become incredibly important sources of information for travellers. Only three in the hospitality and tourism domain reviewed the sentiment analysis tools used in hospitality and tourism and illustrated their utility using TripAdvisor data (Mehraliyev, 2022).

With sentiment analysis, hospitality businesses can keep an eye on and manage their online reputation by tracking and analysing the reviews that are posted on these platforms. Businesses can improve their brand image by addressing negative feedback promptly and highlighting positive experiences to influence potential guest's perceptions.

Businesses can use sentiment analysis to track trends in customer satisfaction across various categories like accommodations, dining, amenities, and customer service (Yussupova, 2014). This helps them identify areas that need improvement and allocate resources to enhance service quality in the areas that matter most to guests. Sentiment analysis is useful for understanding how well a business is doing and for comparing its performance to that of its competitors. When hospitality businesses compare sentiment scores and trends with their competitors, they can figure out what they are good at and where they need to improve compared to their peers. This helps them produce strategies to get ahead in the market. Yes, sentiment analysis is a valuable tool for conducting market research and trend analysis in the hospitality industry (Guo, 2017). Businesses can gain valuable insights by analysing sentiment trends over time and across different demographics or geographic regions.

2.3 Role of Tripadvisor in hospitality industry

Trip advisor is renowned for its vast repository of consumer reviews and ratings about hotels, restaurants, attractions, and other establishments in the hospitality industry. These reviews offer significant feedback to both businesses and potential clients, impacting booking decisions and moulding the reputation of enterprises. Since its launch in 2000, it has garnered more than some 75 million reviews (Chua, 2013). From then travellers across the globe use the Tripadvisor site and app to browse more than 860 million reviews and opinions of 8.7 million accommodations, restaurants, experiences, airlines, and cruises (Trip advisor, n.d.).

Tripadvisor allows businesses to directly interact with customers by replying to reviews, addressing complaints, and offering supplementary information. Around the world and over 115 new contributions posted every minute (Yu, 2017). This also make the trip advisor one of the most trusted and best online travel agencies for the travellers across the globe. This interaction can assist organisations in establishing a positive relationship with clients and highlighting their dedication to ensuring customer satisfaction.

Tripadvisor can be used by hospitality organisations to monitor the results of their competitors, compare their performance to industry benchmarks, and discover areas that require development. This information is extremely helpful for strategic planning and decision-making. A multitude of travellers depend on TripAdvisor to arrange their vacations and determine their booking choices. Across hotel, restaurant, and attraction reviews, more than four out of five (85%) participants report that the reviews they read on TripAdvisor accurately reflect their experience, with 86% agreeing that TripAdvisor makes them feel more confident in their booking decisions (prnewssite, n.d.).

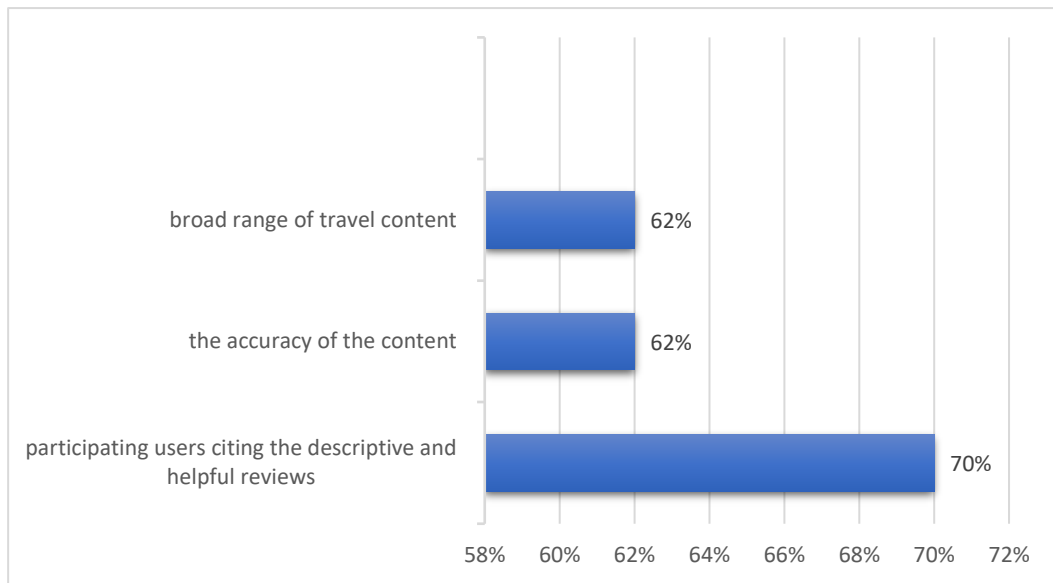


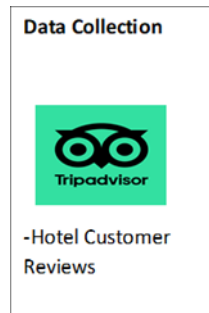
Figure 4: Percentage of people visiting trip advisor platform.

The platform's reviews and ratings are a reliable and impartial source of information for travellers who are looking for genuine and unbiased input on hotels, restaurants, and attractions. TripAdvisor compiles data from a vast number of reviews and ratings, offering useful information on consumer preferences, trends, and up-and-coming travel destinations. Hospitality establishments might use this data to adapt their products to meet changing client requirements.

Overall, Trip Advisor has transformed the hospitality sector by making information more accessible to everyone, giving customers more power, and influencing how businesses operate. The influence of technology keeps growing as advancements are made and consumer preferences change.

Chapter 3

Exploratory Data Analysis



Data Preprocessing

- Missing value
- Data Cleaning



Feature Engineering

- Feature Selection



Model Construction

- Naive Bayes
- Logistic Regression
- Random Forest
- Artificial Neural Networks(ANN)



Result Discussion

- Visual Analysis
- Sentiment Result
- Sentiment Analysis Evaluation

3.1 Ethical Considerations:

The "515k Hotel Reviews Data in Europe" dataset from Kaggle is relatively straightforward in terms of ethical considerations. Since the dataset consists of anonymized hotel reviews without any personally identifiable information, there are minimal privacy concerns associated with it. Additionally, the dataset does not contain sensitive or controversial content that could be considered. As a result, we can use the dataset without significant ethical implications, focusing on analysing and deriving insights from the hotel reviews to understand customer preferences, satisfaction levels, and trends in the hospitality industry with minimal ethical concerns.

3.2 About the Dataset

The dataset for this study consists of a comprehensive collection of hotel reviews sourced from TripAdvisor, one of the largest online travel platforms globally. The dataset encompasses reviews from various destinations across Europe, covering a diverse range of accommodation types. The dataset contains 50,000 records with 17 features covering the period of January 2015 to December 2017.

Key Features of the Dataset:

Review Text: The dataset includes the text of each review, containing valuable insights into guest's experiences, perceptions, and sentiments regarding their stays.

Rating: Each review is accompanied by a numerical rating provided by the reviewer, ranging from 1 to 10 representing the overall satisfaction level with the accommodation.

Reviewer Demographics: Demographic information about the reviewers, such as Nationality, travel purpose.

Hotel Information: Details about the reviewed hotels, including their names, locations, categories (e.g., budget, mid-range, luxury), and amenities, are provided to contextualize the reviews and enable analysis based on hotel characteristics.

Date of Review: The dataset includes timestamps indicating when each review was submitted, allowing for temporal analysis of sentiment trends over time.

Columns	Datatypes	Explanation
Hotel_Address	object	The address of the hotel.
Additional_Number_of_Scoring	int64	There are also some guests who just made a scoring on the service rather than a review. This number indicates how many valid scores without review in there.
Review_Date	object	Date when reviewer posted the corresponding review.
Average_Score	float64	The average score of the hotel based on all the reviews it has received.
Hotel_Name	object	The name of the hotel being reviewed.
Reviewer_Nationality	object	The nationality of the reviewer.
Review_Total_Positive_Word_Counts	int64	The total word count of the positive review.
Total_Number_of_Reviews_Reviewer_Has_Given	int64	The total number of reviews the reviewer has given on TripAdvisor.
Reviewer_Score	float64	The score given by the reviewer to the hotel.
Tags	object	Additional tags associated with the review, indicating specific aspects of the hotel experience (e.g., "family vacation", "business trip").
days_since_review	object	The number of days since the review was written.
Lat	float64	Latitude coordinates of the hotel's location.

lng	float64	Longitude coordinates of the hotel's location.
-----	---------	--

Table 1 : Dataset columns and their types.

3.3 Data Preprocessing

Data preprocessing is a crucial step in preparing the collected data for analysis. Following are the preprocessing techniques which have been conducted in this dataset:

Handling Missing Values: Check for missing values in each column and determine an appropriate strategy for handling them. This may involve employing techniques such as filling missing values with statistical measures like mean, median, mode.

Cleaning Text Data: For textual data such as reviews, execute tasks like removing special characters, punctuation, and unnecessary whitespace. Convert text to lowercase to maintain consistency and eliminate any HTML tags or formatting elements.

Tokenization: Break down the text into individual words or tokens to facilitate further analysis. This process involves splitting the text into smaller units, typically words or phrases, which can then be analysed individually.

Stopword Removal: Eliminate common stopwords (e.g., "the", "and", "is") that carry minimal meaning and are unlikely to contribute significantly to sentiment analysis. Removing stopwords helps reduce noise in the text data and enhances the accuracy of sentiment analysis models.

Stemming or Lemmatization: Apply stemming or lemmatization techniques to reduce words to their base or root form. The words 'user', 'users', 'used', 'using' all can be stemmed to the word 'use'. Stemming helps in reducing the size of the term document matrix hereby reducing the dimension of data (Mhatre, 2017, July).

Feature Engineering: Generate new features or extract additional insights from the existing data. For instance, compute the length of reviews in terms of word count or create binary indicators for the presence of positive or negative sentiment keywords.

Encoding Categorical Variables: Transform categorical variables like reviewer nationality or hotel category into numerical format using methods like one-hot encoding.

Normalization: Scale numerical features to a standardized range to prevent certain features from dominating the analysis. Common normalization techniques include min-max scaling.

By executing these preprocessing steps, the dataset is refined, standardized, and formatted into a suitable structure for analysis, ensuring the accuracy and reliability of subsequent analytical procedures.

3.4 Data Analysis

Before conducting data analysis, we performed data preprocessing steps including handling missing values, data cleaning, and data transformation, as necessary. This ensures the data is ready for analysis and reduces the likelihood of erroneous conclusions.

We started by computing descriptive statistics such as mean, median, standard deviation, and quartiles for numerical variables. This provided a summary of the central tendency and spread of the data.

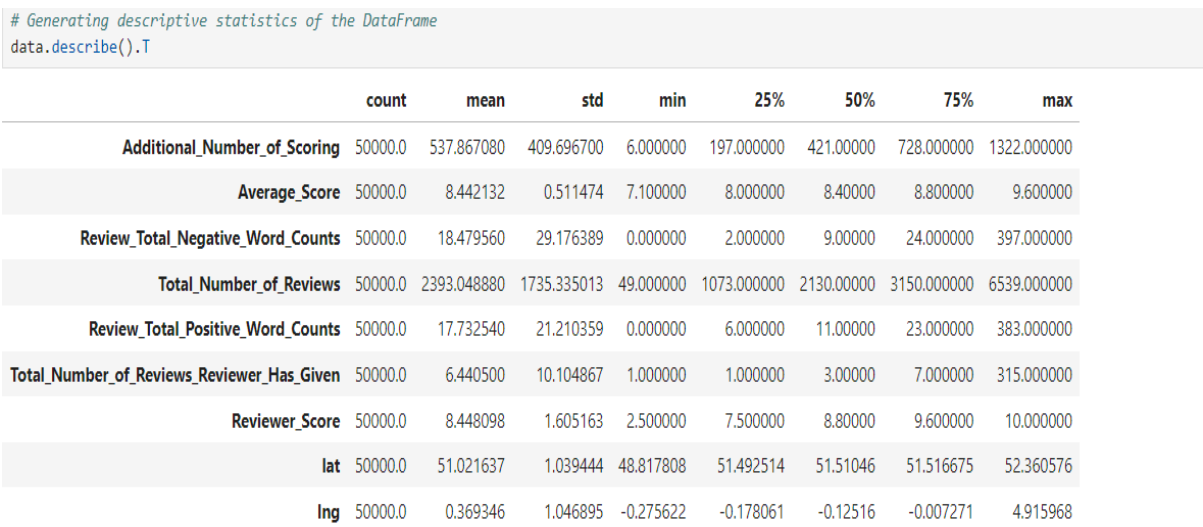


Figure 5: Descriptive statistics of the data.

3.4.1 Findings and Insights

We utilized the longitude and latitude columns from the dataset to generate a visual representation of the geographical locations of top-rated hotels. The resulting map highlights

the of high-rated hotels, primarily concentrated in specific regions. Notably, our analysis revealed significant concentrations of top-rated hotels in both Paris and London.

In Paris, these high-rated hotels appear to cluster around popular tourist destinations such as the Eiffel Tower, Champs-Élysées, and the Louvre Museum. Meanwhile, in London, the top-rated hotels are concentrated around landmarks like Buckingham Palace, the Tower of London, and Hyde Park.

According to the World Tourism Cities Development, London is ranked third in the world, and ranked first in Europe and first in popularity (Chang, 2023).

Drop off locations of hotels

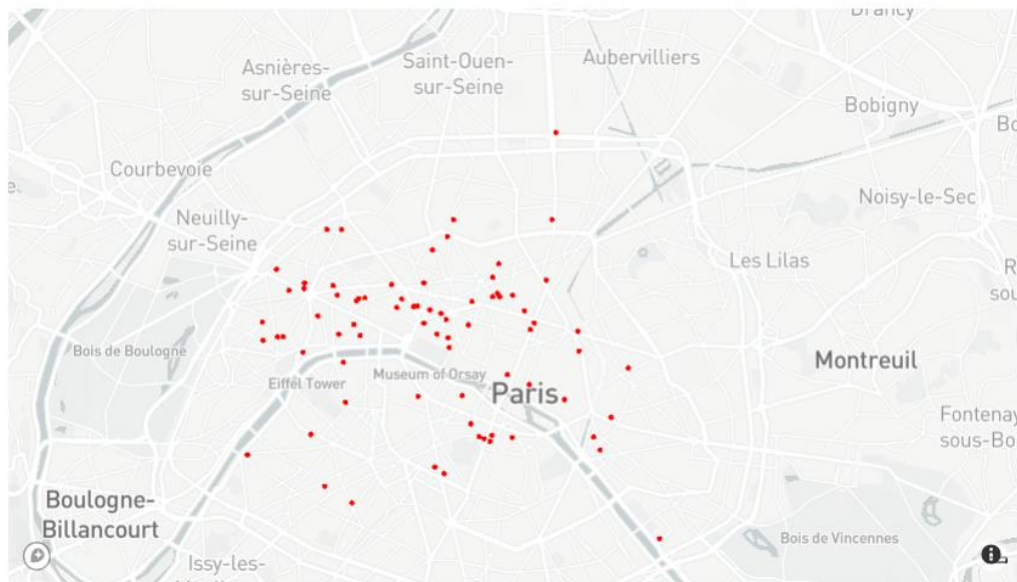


Figure 6 : Hotel location in Paris

Drop off locations of hotels

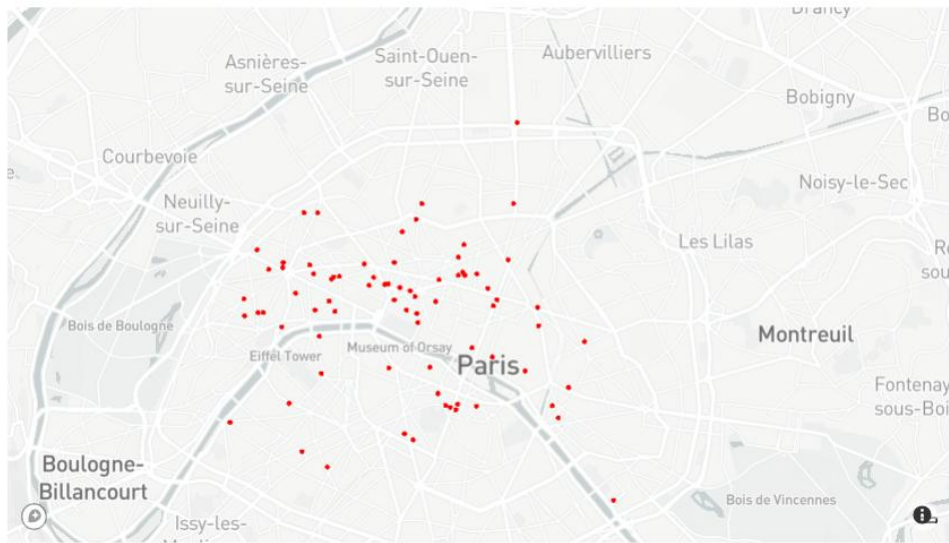


Figure 6: Hotel location in Paris.

This geographical insight provides valuable information for travellers seeking accommodation in these cities. By identifying the areas with a high density of top-rated hotels, travellers can make more informed decisions when selecting accommodations based on their preferences and proximity to key attractions.

Additionally, this analysis offers insights for hoteliers and stakeholders in the hospitality industry. Understanding the geographic distribution of top-rated hotels allows them to assess competition, strategically position their properties, and tailor marketing efforts to target high-traffic areas frequented by travellers.

The images provided highlight the top 10 hotels based on the highest volume of reviews submitted by travellers. These hotels have substantial attention and feedback from guests, indicating their popularity and significance within the hospitality industry.



Figure 7: Top 10 Hotels

For travellers, the list of top-reviewed hotels offers a valuable resource for trip planning, presenting a curated selection of accommodations that have received positive feedback from numerous guests. This suggests the likelihood of exceptional overall experiences at these establishments.

The dataset provided includes details regarding the nationality of reviewers, alongside the identification of the top 10 nationalities represented among reviewers, Where United Kingdom takes the top spot with more than 20k plus reviewers followed by united states. Additionally, it encompasses the distribution of scores given by reviewers, offering valuable insights into their assessments.

Understanding the nationality of reviewers is essential for grasping the demographic composition of individuals providing feedback.

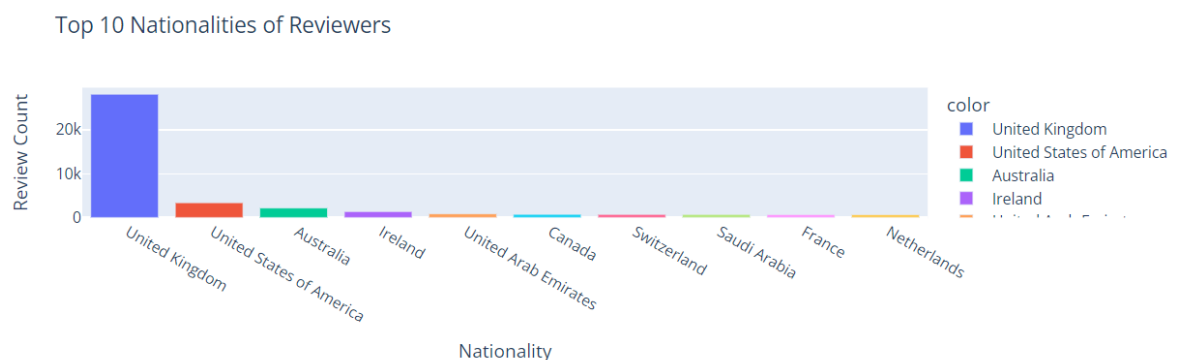


Figure 8: Reviewers Nationality.

Moreover, the identification of the top 10 nationalities among reviewer's sheds light on the diversity of guests contributing feedback. This understanding can assist hoteliers in accommodating the needs and expectations of various cultural groups, thereby enhancing overall guest satisfaction and experience.

The breakdown of scores provided offers a comprehensive overview of guest satisfaction levels. By analysing this distribution, hoteliers can pinpoint both areas of strength and opportunities for improvement, guiding efforts to enhance service quality and guest satisfaction effectively.

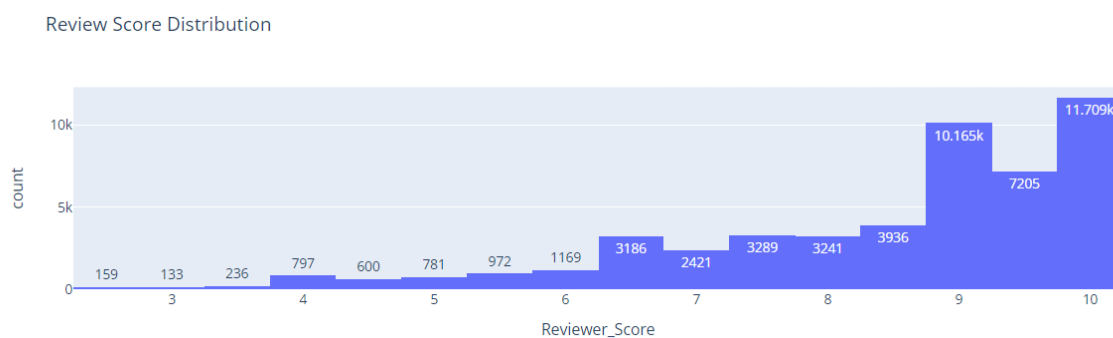


Figure 9: Reviewers Score

The positive and negative word clouds visually depict the frequent terms used by travellers in their reviews, categorized as positive or negative sentiments.

Positive Word Cloud:

The positive word cloud illustrates terms commonly employed by travellers to convey satisfaction, enjoyment, and positive experiences during their hotel stays. These terms may encompass "excellent," "helpful," "clean," "friendly," "great," and "lovely," among others. The prominence of these positive descriptors indicates the prevalence of positive sentiments in the reviews, highlighting aspects of the hotel experience that guests found particularly pleasing.

Word Cloud of Positive Reviews

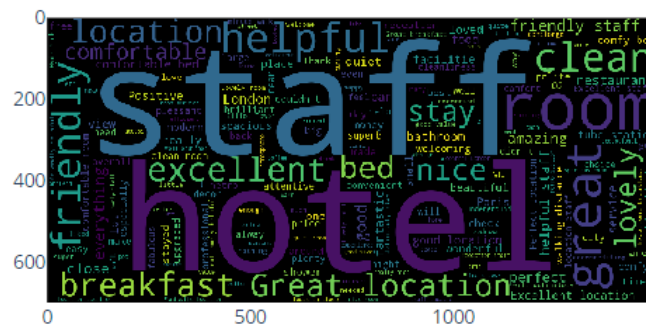


Figure 10: Positive review word cloud.

Negative Word Cloud:

In contrast, the negative word cloud exhibits terms frequently utilized by travellers to express dissatisfaction, disappointment, or negative encounters during their hotel stays. These terms may include "dirty," "small," "disappointing breakfast," "money," "poor service," and "bad," among others. The prominence of these negative descriptors suggests areas where guests encountered deficiencies or experienced issues that diminished their overall satisfaction with the hotel.

Word Cloud of Negative Reviews

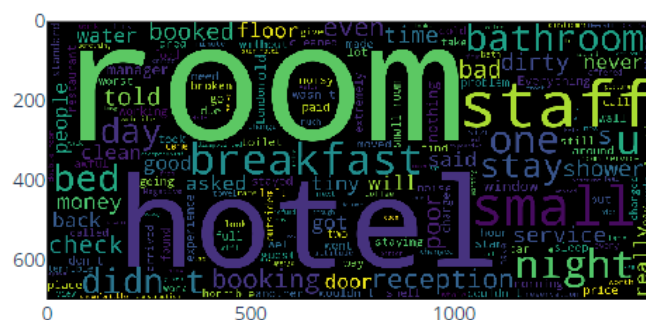


Figure 11: Negative review word cloud.

Analysing of both word clouds provides valuable insights into the aspects of the hotel experience that guests frequently commend or criticize. Hotel managers can use this data to identify strengths to emphasize and areas for improvement to address, thereby improving guest satisfaction and the overall quality of their properties. Similarly, travellers can use these word clouds to guide their decision-making process when choosing accommodations, considering both positive and negative feedback provided by previous guests.

Chapter 4

Model Building

4.1 Train Test Split

The train-test split is a crucial step in machine learning because it allows us to evaluate how well a model performs. The reason we split the dataset into training and testing sets is to evaluate how well the model performs on data it has not seen before. The model is first trained using the training set and then it is evaluated using the testing set. Usually, the dataset is split into two parts: a training set and a testing set. Splitting ratios commonly used are 60:40, 70:30, 80:20, and 90:10 (Muraina, 2022) . The split ratio used to allocate data for training, while the remaining portion is reserved for testing. We use the 80:20 ratio in our case. Before splitting the data, it is important to randomise it. This ensures that the distribution of classes or labels is similar in both the training and testing sets. The training set is used to train the model by adjusting its parameters. On the other hand, the testing set is used to evaluate the model's performance and estimate its generalisation error. Sometimes, when there is a limited dataset, we can use cross-validation techniques like k-fold cross-validation. Here is how it works: we divide the data into several smaller groups called folds. Then, we train the model using different combinations of these folds. Finally, we average the results to get a more reliable estimate of the model's performance. The k error estimator depends on two factors: the training set and the partition into folds (Rodriguez, 2009). In machine learning, the train-test split is an important practice to make sure that the model's performance is accurate and can work well with new data.

4.2 Natural Language Processing(NLP)

In our study we are using two well-known NLP techniques they are Bag of words(BOW) and TF-IDF representation for converting the text data to numerical which is further used for the machine learning model. By compare the technique we conclude which model more suitable for our analysis.

4.2.1 Bag-of-words representation

Bow representation will remove accents from characters using Unicode encoding. It specifies that the analyser will be based on words. Stop words are commonly used words (e.g., "the", "is", "in") that often do not carry significant meaning so those stop words will be removed from the text.

The words are used for the training data and transforms it into a Bag-of-Words representation. It learns the vocabulary from the training data and encodes each review into a sparse matrix where each row corresponds to a review, and each column corresponds to a word in the vocabulary, with values representing the frequency of each word in the respective review. The BOW model assigns a vector to a document as $d=(x_1, x_2, \dots, x_l)$, where x_i denotes the normalized number of occurrence of the i th basis term and l is the size of the collection of basis terms (Mao, April 2018).

We have created the Naive Bayes, Logistic Regression, Random Forest, and Artificial Neural Network model using the BOW representation.

Models	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.79	0.80	0.80	0.80
Naïve Bayes	0.79	0.81	0.76	0.78
XG Boost	0.80	0.80	0.79	0.80
ANN	0.80			

Table 2 : Model results with BOW representation.

4.2.2 TF- IDF Representation

TF-IDF stands for Term Frequency-Inverse Document Frequency. It is a numerical statistic used in natural language processing and information retrieval to evaluate the importance of a word is to a document in a collection or corpus. TF-IDF combines two important concepts:

Term Frequency : Which measures how frequently a term occurs in a document (Vishwakarma, 2015).

a Term Frequency (t)=Number of times term t appears in document

Inverse Document Frequency : Which measures how important a term is. While computing TF, all terms are considered equally important (Vishwakarma, 2015).

Inverse Document Frequency (t)=LogN/Nt

We have created the Naive Bayes, Logistic Regression, Random Forest, and Artificial Neural Network model using the TF-IDF representation.

Models	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.80	0.79	0.83	0.81
Naïve Bayes	0.80	0.79	0.83	0.81
XG Boost	0.77	0.77	0.78	0.78
ANN	0.81			

Table 3: Model results with TF- IDF representation.

Chapter 5

Hyperparameter Tuning

5.1 Parameter Tuning

Hyper parameter tuning is the key process while building a ML or neural network model. It helps the model to learn and use all the estimator to give the best results. In other way the application of effective hyperparameter optimisation techniques can simplify the process of identifying the best hyperparameters for the models. The HPO consists of four main components: First, we need an estimator that can be a regressor or any classifier with one or more objective functions. Second, we need a search space. Third, we need an optimisation method to find the best combinations. And fourth, we need a function to compare the effectiveness of different hyperparameter configurations (Timmer, 2021).

5.1.1 Logistic Regression

While building the logistic regression model, the model accuracy was not but to the level. While performing the hyper parameter tuning the iteration level reached its maximum and some warning messages were popped up by the jupyter notebook. Then by adding the parameters suggested by the scikit learn documents, the model performed well with the accuracy of 80%.

```
# Initializing SVM classifier with probability estimates enabled
Logistic_reg = LogisticRegression()

# Training the SVM classifier on the Bag-of-words representation of the training data
Logistic_reg.fit(X_train_bow, y_train)

# Making predictions on the TF-IDF representation of the testing data
y_pred = Logistic_reg.predict(X_test_bow)
```

C:\Users\Diraj\Anaconda3\lib\site-packages\sklearn\linear_model_logistic.py:460: ConvergenceWarning:
lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

Figure 12: Logistic Regression warning message.

The solvers implemented in the class `LogisticRegression` are "lbfgs", "liblinear", "newton-cg", "newton-cholesky", "sag" and "saga":

The following table summarizes the penalties supported by each solver:

Penalties	Solvers					
	'lbfgs'	'liblinear'	'newton-cg'	'newton-cholesky'	'sag'	'saga'
Multinomial + L2 penalty	yes	no	yes	no	yes	yes
OVR + L2 penalty	yes	yes	yes	yes	yes	yes
Multinomial + L1 penalty	no	no	no	no	no	yes
OVR + L1 penalty	no	yes	no	no	no	yes
Elastic-Net	no	no	no	no	no	yes
No penalty ('none')	yes	no	yes	yes	yes	yes
Behaviors						
Penalize the intercept (bad)	no	yes	no	no	no	no
Faster for large datasets	no	no	no	no	yes	yes
Robust to unscaled datasets	yes	yes	yes	yes	no	no

The "lbfgs" solver is used by default for its robustness. For large datasets the "saga" solver is usually faster. For large dataset, you may also consider using `SGDClassifier` with `loss="log_loss"`, which might be even faster but requires more tuning.

Figure 13: Summary of the solver.

```
LR = LogisticRegression(solver='sag', max_iter=1000)

model_predict(LR, X_train_bow, y_train, X_test_bow, y_test)
```

Accuracy: 0.7984954325631382

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.80	0.80	1864
1	0.80	0.80	0.80	1858
accuracy			0.80	3722
macro avg	0.80	0.80	0.80	3722
weighted avg	0.80	0.80	0.80	3722

Figure 14: Result after Tuning.

5.1.2 XG Boost

In the first iteration of the xg boost model, the predict was 79% after while tuning the parameters by changing the `n_estimators` and `depth`, the accuracy increased and the final predicted was 79%. Below are images for the results with different parameters.

```
xg = xgb.XGBClassifier()

model_predict(xg, X_train_bow, y_train, X_test_bow, y_test)
```

Accuracy: 0.7756582482536271

Classification Report:

	precision	recall	f1-score	support
0	0.78	0.77	0.77	1864
1	0.77	0.78	0.78	1858
accuracy			0.78	3722
macro avg	0.78	0.78	0.78	3722
weighted avg	0.78	0.78	0.78	3722

Figure 15: Result before tuning.

```
xg = xgb.XGBClassifier(n_estimators= 300, max_depth=3)

model_predict(xg, X_train_bow, y_train, X_test_bow, y_test)
```

Accuracy: 0.7888232133261688

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.78	0.79	1864
1	0.78	0.80	0.79	1858
accuracy			0.79	3722
macro avg	0.79	0.79	0.79	3722
weighted avg	0.79	0.79	0.79	3722

Figure 16: Result after tuning.

```
xg = xgb.XGBClassifier(n_estimators= 700, max_depth=5)

model_predict(xg, X_train_bow, y_train, X_test_bow, y_test)
```

Accuracy: 0.7941966684578183

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.79	0.79	1864
1	0.79	0.80	0.79	1858
accuracy			0.79	3722
macro avg	0.79	0.79	0.79	3722
weighted avg	0.79	0.79	0.79	3722

Figure 17: Result after changing the parameter.

```
xg = xgb.XGBClassifier(n_estimators= 500, max_depth=5)

model_predict(xg, X_train_bow, y_train, X_test_bow, y_test)
```

Accuracy: 0.7963460505104782

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.79	0.80	1864
1	0.79	0.80	0.80	1858
accuracy			0.80	3722
macro avg	0.80	0.80	0.80	3722
weighted avg	0.80	0.80	0.80	3722

Figure 18: Result.

5.1.3 Artificial Neural Network

Below is the initial architecture of an artificial neural network (ANN) consisting of three dense layers. We aim to optimize its performance by tuning the activation functions, reducing the number of layers, reducing the epoch, and incorporating other activation functions.

We will experiment with different activation functions such as sigmoid, SoftMax, and ReLU to determine their impact on model performance. We will assess whether reducing the number of layers while maintaining model complexity improves computational efficiency

without compromising accuracy. Finally, we end up with the softmax function and two dense layers.

```
1 def build_model_1():
2     model = Sequential()
3     model.add(Dense(256, activation='relu', input_shape=(reordered_sparse_tensor.shape[1],)))
4     model.add(Dropout(0.25))
5     model.add(Dense(128, activation='relu'))
6     model.add(Dropout(0.25))
7     model.add(Dense(1, activation='sigmoid'))
8
9     model.compile(loss='binary_crossentropy',
10                  metrics=['accuracy'],
11                  optimizer=Adam(learning_rate=0.001))
12     return model

1 model_1.evaluate(reordered_sparse_tensor_t, y_test)
2
```

196/196 [=====] - 1s 4ms/step - loss: 1.0918 - accuracy: 0.7677
[1.0918327569961548, 0.7676573991775513]

Figure 19 : Initial ANN model and results.

```
1 def build_model_1():
2     model = Sequential()
3     model.add(Dense(256, activation='sigmoid', input_shape=(reordered_tensor.shape[1],)))
4     model.add(Dropout(0.25))
5     model.add(Dense(1, activation='sigmoid'))
6
7     model.compile(loss='binary_crossentropy',
8                  metrics=['accuracy'],
9                  optimizer=Adam(learning_rate=0.001))
10    return model

1 model_1.evaluate(reordered_tensor_t, y_test)
```

146/146 [=====] - 0s 2ms/step - loss: 0.4530 - accuracy: 0.7984
[0.4530254304409027, 0.7983663082122803]

Figure 20: ANN model with changing the activation function.

```
def build_model_1():
    model = Sequential()
    model.add(Dense(256, activation='sigmoid', input_shape=(reordered_tensor.shape[1],)))
    model.add(Dropout(0.25))
    model.add(Dense(128, activation='sigmoid', input_shape=(reordered_tensor.shape[1],)))
    model.add(Dropout(0.25))
    model.add(Dense(1, activation='sigmoid'))

    model.compile(loss='binary_crossentropy',
                  metrics=['accuracy'],
                  optimizer=Adam(learning_rate=0.001))
    return model

1 model_1.evaluate(reordered_tensor_t, y_test)

146/146 [=====] - 0s 2ms/step - loss: 0.4851 - accuracy: 0.7896
[0.4851118326187134, 0.7895528674125671]
```

Figure 21: ANN model with add on layers.

We have identified that training the model for 5 epochs produces superior outcomes in comparison to training it for 20 epochs. This observation indicates that the model is likely to experience overfitting if it is trained for a longer period.

Overfitting arises when the model acquires the ability to memorise the training data instead of extracting general patterns from it, leading to poor performance on new, unknown data. By decreasing the number of epochs from 20 to 5, it is possible that you have minimised overfitting and attained improved generalisation.

```
1 def build_model_1():
2     model = Sequential()
3     model.add(Dense(256, activation='softmax', input_shape=(reordered_tensor.shape[1],)))
4     model.add(Dropout(0.25))
5     model.add(Dense(1, activation='sigmoid'))
6
7     model.compile(loss='binary_crossentropy',
8                   metrics=['accuracy'],
9                   optimizer=Adam(learning_rate=0.001))
10    return model

1 model_1 = build_model_1()
2 history = model_1.fit(
3     reordered_tensor, y_train,
4     epochs=5,
5     batch_size=16,
6     validation_data=(reordered_tensor_t, y_test),
7     verbose=1,
8     shuffle=True
9 )

1 model_1.evaluate(reordered_tensor_t, y_test)

146/146 [=====] - 0s 2ms/step - loss: 0.4395 - accuracy: 0.8061
[0.4395480751991272, 0.8061048984527588]
```

Figure 22: Final ANN model with result.

Chapter 6

Conclusion

6.1 Results

Conclusions based on the accuracy scores supplied for several machine learning models trained on both Bag-of-Words (BoW) and TF-IDF representations.

6.1.1 BoW Representation

Logistic Regression:

Accuracy	Precision	Recall	F1 Score
0.81	0.82	0.79	0.80

Table 4: Logistic Regression performance metrics.

With an accuracy of 0.81, the model correctly predicts 81% of the observations. With a precision of 0.82, 82% of the events that the model predicted as true positives are positive. Recall of 0.79 means that 79% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.80 indicates that recall and precision are in balance and both metrics are at a relatively high level.

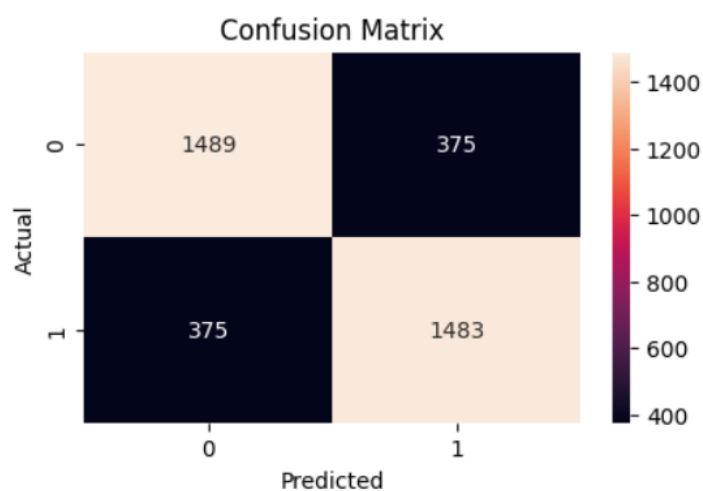


Figure 23: Confusion matrix for logistic regression.

Naïve Bayes:

Accuracy	Precision	Recall	F1 Score
0.80	0.82	0.76	0.79

Table 5 : Naïve Bayes performance metrics.

With an accuracy of 0.80, the model correctly predicts 80% of the observations. With a precision of 0.82, 82% of the events that the model predicted as true positives are positive. Recall of 0.76 means that 76% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.79 indicates that recall and precision are in balance and both metrics are at a relatively high level.

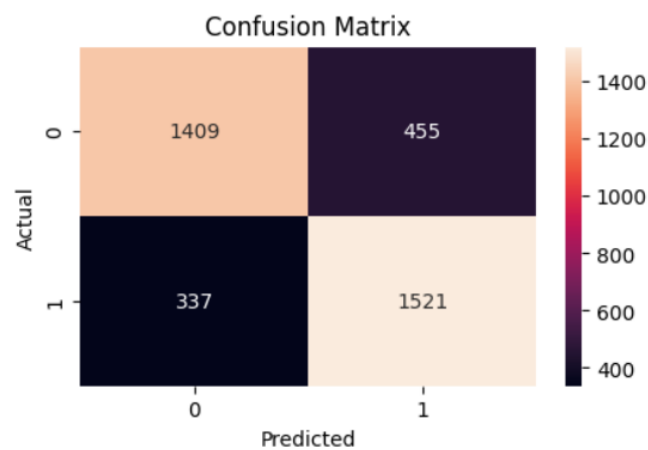


Figure 24: Confusion matrix for naive bayes.

XG Boost:

Accuracy	Precision	Recall	F1 Score
0.80	0.81	0.78	0.80

Table 6 : XG boost performance metrics.

With an accuracy of 0.80, the model correctly predicts 80% of the observations. With a precision of 0.81, 81% of the events that the model predicted as true positives are positive. Recall of 0.78 means that 78% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.80 indicates that recall and precision are in balance and both metrics are at a relatively high level.

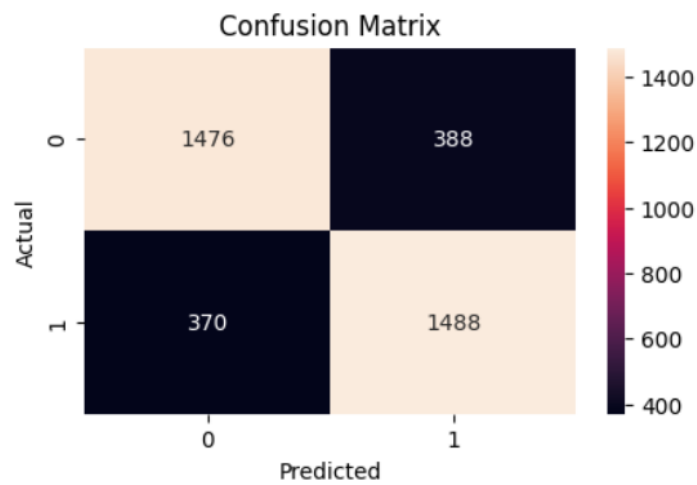


Figure 25: Confusion matrix for Xg boost.

ANN:

Train loss	Val loss	Train accuracy	Val accuracy
0.1594	0.4197	0.9683	0.8170

Table 7: ANN result.

Train loss indicates how well the model fits with the training data. In this case, the train loss is 0.1594. The val loss indicates how well the model is generalizing to unseen data. A val loss of 0.4197 suggests that the model's performance on the validation set is slightly worse than

on the training set, which is normal. The Train Accuracy indicates the proportion of correctly classified instances in the training set. A train accuracy of 0.9683 means that the model correctly classifies 96.8% of the training set instances. The val accuracy indicates the proportion of correctly classified instances in the validation set. A val accuracy of 0.817 suggests that the model correctly classifies 81.7% of the validation set instances.

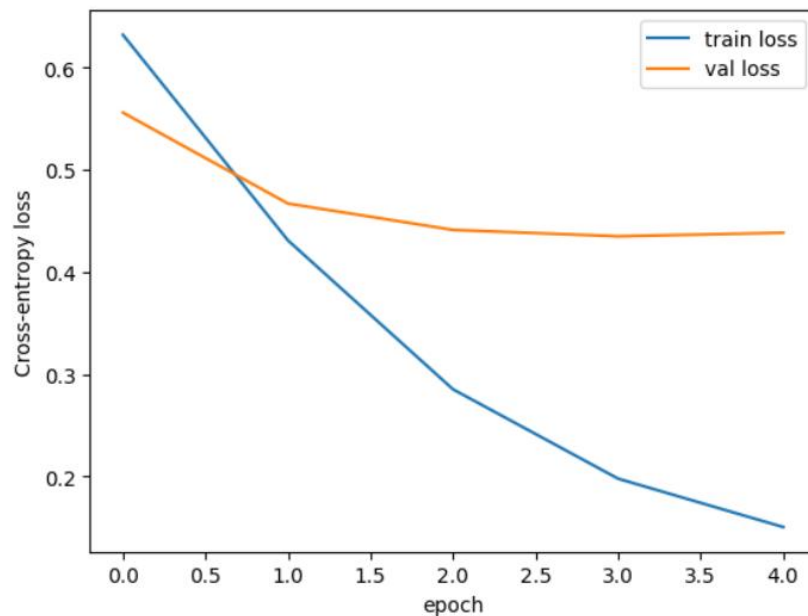


Figure 26: Training and validation loss.

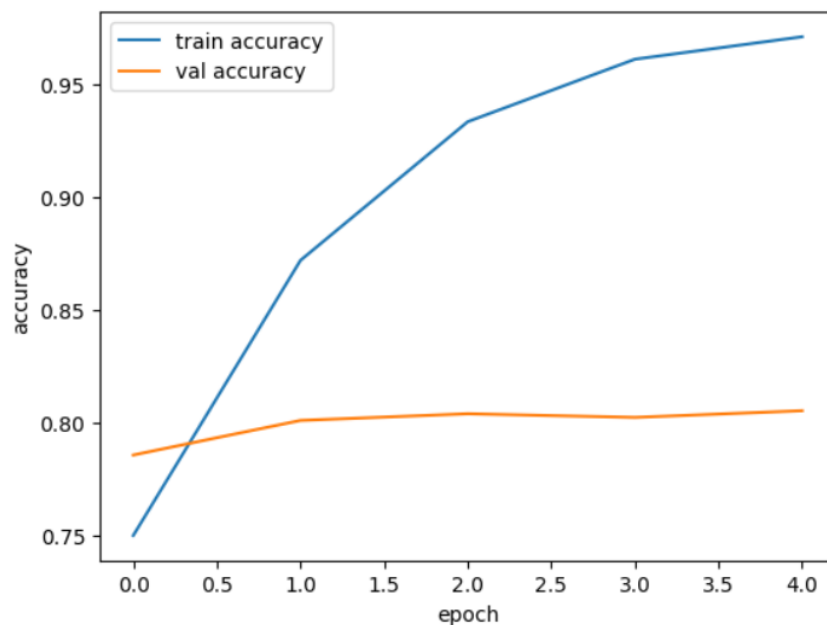


Figure 27: Training and validation accuracy.

6.1.2 TF-IDF Representation

Logistic Regression:

Accuracy	Precision	Recall	F1 Score
0.82	0.81	0.83	0.82

Table 8: Logistic Regression performance metrics.

With an accuracy of 0.82, the model correctly predicts 82% of the observations. With a precision of 0.81, 81% of the events that the model predicted as true positives are positive. Recall of 0.83 means that 83% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.82 indicates that recall and precision are in balance and both metrics are at a relatively high level.

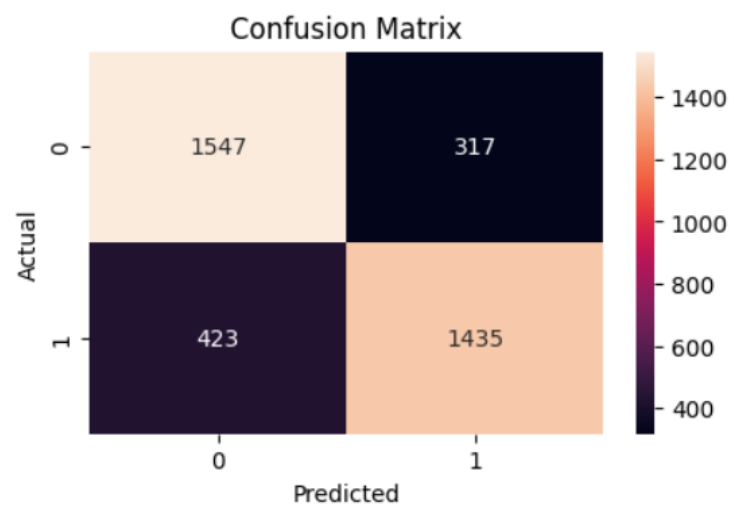


Figure 28: Confusion matrix for tf-idf logistic regression.

Naïve Bayes:

Accuracy	Precision	Recall	F1 Score
0.81	0.80	0.83	0.82

Table 9 : Naïve bayes performance metrics.

With an accuracy of 0.81, the model correctly predicts 81% of the observations. With a precision of 0.80, 80% of the events that the model predicted as true positives are positive. Recall of 0.83 means that 83% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.82 indicates that recall and precision are in balance and both metrics are at a relatively high level.

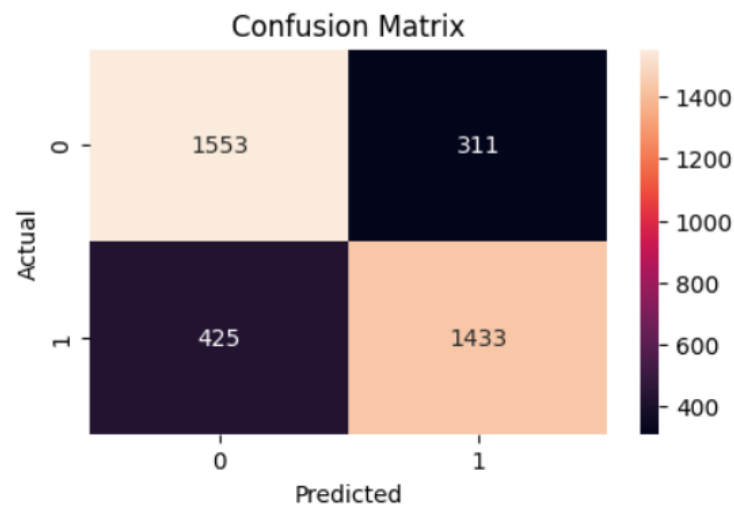


Figure 29: Confusion matrix for tf-idf naive bayes.

XG Boost:

Accuracy	Precision	Recall	F1 Score
0.79	0.77	0.79	0.79

Table 10 : Xg boost performance metrics.

With an accuracy of 0.79, the model correctly predicts 79% of the observations. With a precision of 0.77, 77% of the events that the model predicted as true positives are positive. Recall of 0.79 means that 79% of real positive events are properly identified by the model. $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$ is the calculation. An F1 score of 0.79 indicates that recall and precision are in balance and both metrics are at a relatively high level, and these are very low compared to the other models.

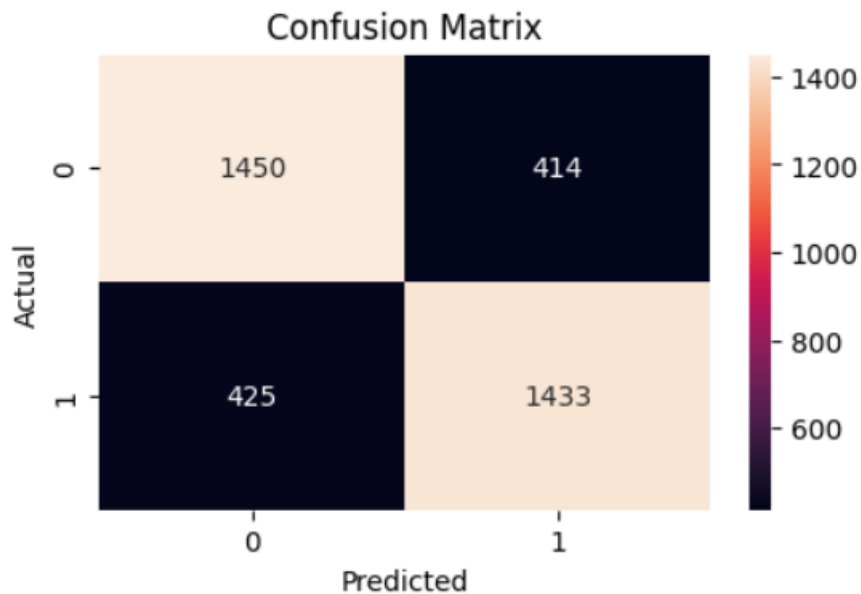


Figure 30 :Confusion matrix for tf-idf xg boost.

ANN:

Train loss	Val loss	Train accuracy	Val accuracy
0.3874	0.4232	0.8597	0.8152

Table 11 : ANN performance result.

Train loss indicates how well the model fits with the training data. In this case, the train loss is 0.3874. The val loss indicates how well the model is generalizing to unseen data. A val loss of 0.4232 suggests that the model's performance on the validation set is slightly worse than on the training set, which is normal. The Train Accuracy indicates the proportion of correctly classified instances in the training set. A train accuracy of 0.8597 means that the model correctly classifies 85.9% of the training set instances. The val accuracy indicates the proportion of correctly classified instances in the validation set. A val accuracy of 0.8152 suggests that the model correctly classifies 81.52% of the validation set instances.

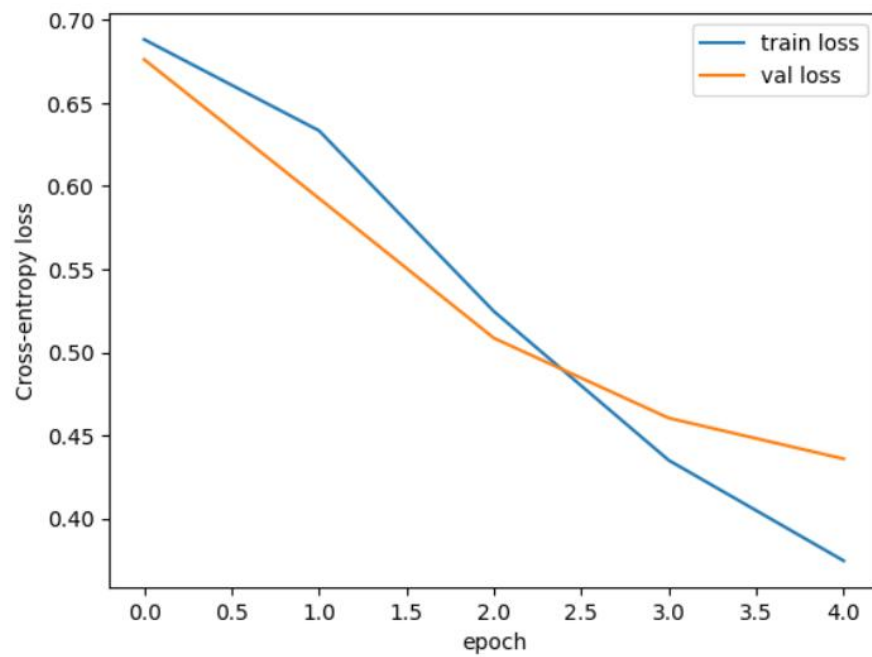


Figure 31: Training and validation loss.

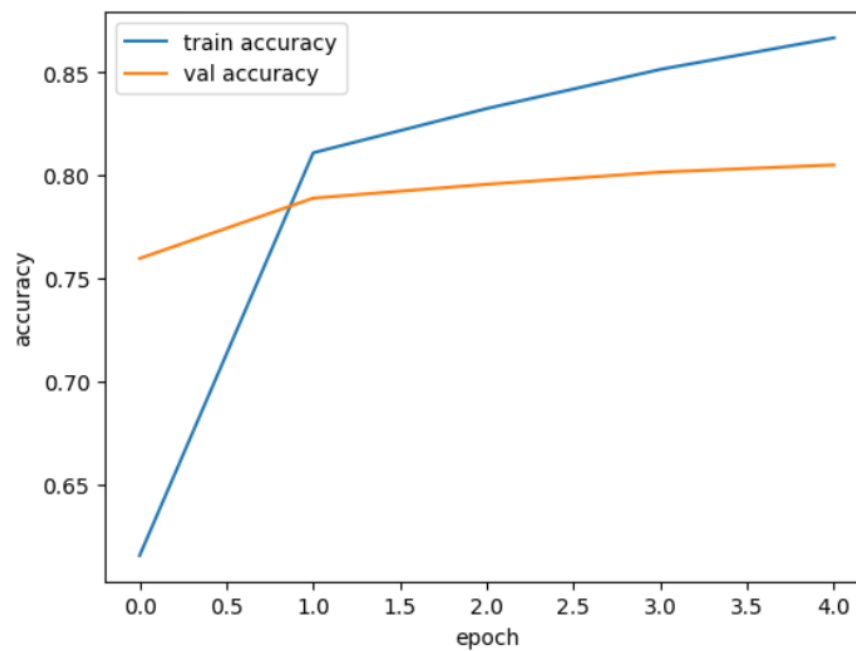


Figure 32: Training and validation accuracy.

6.2 Comparison

In both representations, the Artificial Neural Network (ANN) consistently outperforms the other methods, attaining the greatest accuracy scores. Logistic Regression and Naive Bayes exhibit comparable performance in both forms, with minor deviations in accuracy. The TD-IDF representations performs was fast in terms of the CPU cores used and the time taken for the whole code to run, and any time spent waiting for resources like I/O operations or external processes. Below are the observations for the performance of both the representation.

Model	CPU times	Wall time
Logistic Regression	14s	24s
Naïve Bayes	188ms	298ms
XG Boost	9min 45s	2min 26s
ANN	5min 44s	2min 30s

Table 12: Bow representation performance results.

Model	CPU times	Wall time
Logistic Regression	422ms	517s
Naïve Bayes	156ms	213ms
XG Boost	2min 7s	19.3s
ANN	1min 30s	1min 52s

Table 13 : Tf-idf representation performance results.

For BoW representation, both training and validation losses are less compared to TF-IDF representation and TF-IDF representation, the training loss is lower, indicating better generalization compared to TF-IDF. The training accuracy is higher for BoW representation compared to TF-IDF representation. This indicates that BoW has better performance with the training data and TF-IDF representation may generalize better performance with the unseen data.

Overall, the ANN model with BoW representation seems to perform better in terms of training loss and accuracy, indicating better model performance.

6.3 Future Works

Our future works mainly focuses on addressing the following two challenges we have faced during our project.

Complexity of the Project:

Complexity of the project which involves scope, execution of the code, scale, and the size of the dataset. The scope of the project encompasses not only data collection and preprocessing but also the selection and implementation of appropriate machine learning and natural language processing techniques. Scaling the project to handle a large dataset requires optimization of algorithms and code, as well as consideration of computational resources. Since the dataset records were roughly 0.2 M it was challenging to extract them in the Jupyter environment and work with NLP and machine learning techniques. This is a result of the machine having extremely little RAM and GPU core memory. Even with Google Colab's basic free edition, the ML and neural network model performed poorly. Working with a dataset of this size may require strategies such as data sampling, batch processing, or using cloud-based resources to manage computational demands.

Novelty of Methods:

The novelty of methods in a project to the use of innovative or original approaches, techniques which were widely used for the text classification to improve the accuracy. Implementing NLP and machine learning techniques involves writing and debugging code, which can be complex, especially when dealing with large datasets. Ensuring the code runs efficiently and effectively requires careful attention to detail and thorough testing. Through the research from the journals, we can understand that the deep learning model has performed well. Deep Belief Neural (DBN), Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN), these three main types of deep learning architectures, are largely explored to handle various classification tasks (Zulqarnain, 2020). Also, we have BERT (Bidirectional Encoder Representations from Transformers) able to cope with NLP tasks such as supervised text classification without human supervision. Its flexibility to cope with any corpus delivering great results has made this approach very popular in academia and industry techniques for better results (González-Carvajal, 2020). And we can explore more on Word embeddings, such as GloVe, and FastText. By exploring these additional methods and techniques, researchers can further improve the accuracy of text classification models and overcome the limitations of handling large datasets.

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